Thermal Applications of xGnP® Graphene Nanoplatelets

Percy Chinoy

Hiroyuki Fukushima

Thomas Ritch

Jeffri Narendra

Liya Wang

November 19, 2015

Graphene and 2D Materials Conference

Santa Clara CA

Overview

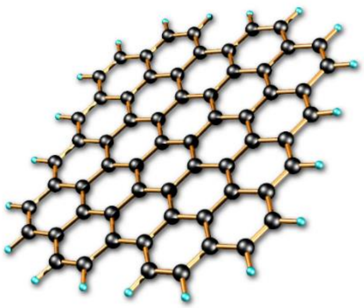
- Introduction – XG Sciences and Graphene Nanoplatelets
- Thermal Interface Materials
- Heat Spreaders
- Conductive Inks and Coatings
- Summary

XG Sciences

- Founded in 2006 in Lansing, MI
- 850+ Customers in 40+ countries
- World class Strategic Partners and Licensees
- Significant IP: patents, trade secrets and know-how
- 25,000 ft^2 Factory (≈2,300 m^2), >100 Tons capacity
- Commercial Supplier of…
  - Bulk xGnP® Graphene Nanoplatelets
    - Multiple grades and sizes (<1 µm to >50 µm)
    - Dry powders or Dispersions
  - Value-Added Product Platforms
    - XG SiG™ Silicon Graphene Composites for LiB Anodes
    - XG Leaf® Sheet product for Thermal and Electrical functionality
    - XG TIM™ Thermal Interface Materials
    - Inks & Coatings

Graphene and Nano Carbon Materials

Single-layer Graphene

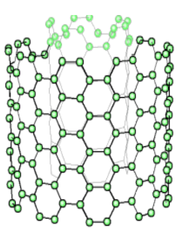


http://www.zeitnews.org

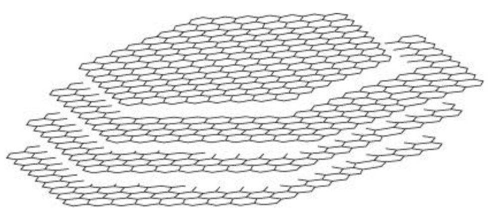
- One-atom-thick layer of carbon atoms in an sp^2 bonded, honeycomb-shaped, 2-dimensional network
  - High thermal conductivity
  - High electric conductivity
  - Excellent mechanical strength and flexibility
- Typically produced by vapor deposition on substrates – challenges in cost, scale up, application flexibility

---

Carbon Nanotube



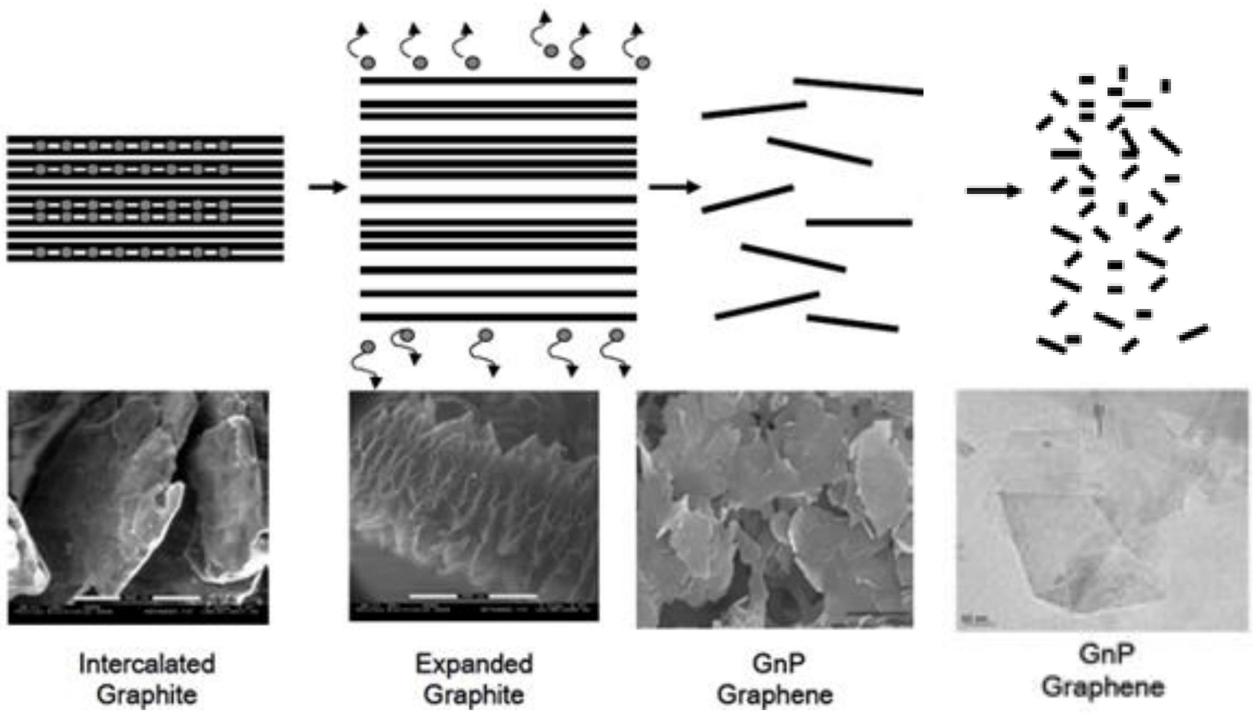
xGnP® Graphene Nanoplatelet

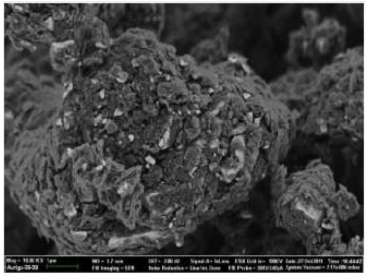


XG Sciences uses low-cost, **non-oxidizing processes** to produce **high purity, low defect, graphene nanoplatelets** (≈2 to 20nm thickness) whose size, shape & edge chemistry can provide **customized properties**

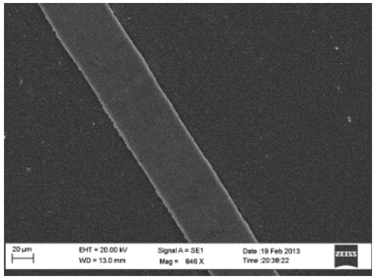
Grade H	50 – 80 m^2/g	5, 15 or 25 μm diameters
Grade M	120 – 150 m^2/g	5, 15 or 25 μm diameters
Grade C	300 – 750 m^2/g	small diameter particles

Scalable Manufacturing Processes for xGnP® Graphene Nanoplatelets and Value-added Products

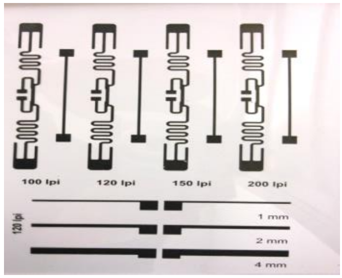




XG SiG™



XG Leaf®

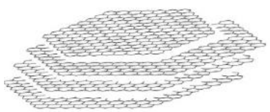


XG Inks™

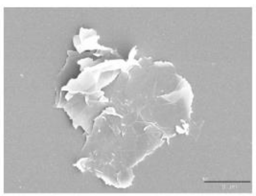
Proprietary scalable manufacturing processes for Intercalation, Exfoliation, Mixing and Calendering

xGnP® enables a broad range of Applications

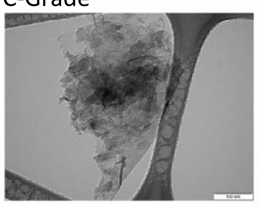
MATERIAL — ATTRIBUTE — PROPERTY — FUNCTION — APPLICATION



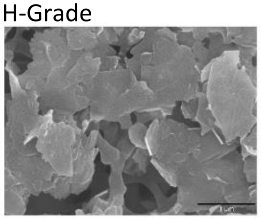
M-Grade



C-Grade



H-Grade



Attributes:

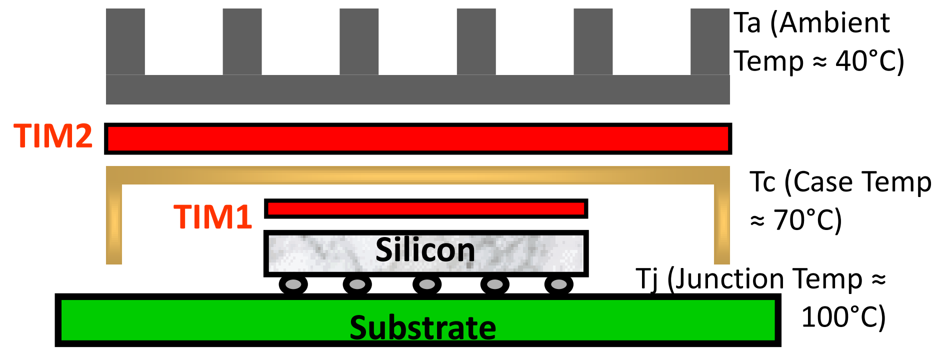
- Platelet Size and Thickness
- Surface Chemistry
- Aggregate Morphology
- Pore Size and Distribution

PROPERTY	FUNCTION	APPLICATION
Thermal	Thermal Conductor Heat Management	Automotive, LED Lighting, LED & QD Display, Electronics
Electrical	Conductor Controlled Resistor	RFID, Smart Tags, Printed Circuits, Touch Panels, Printers
Mechanical	Modulus Strength Scratch Resistance	Automotive Exterior Parts, Electronics Bodies
Barrier	Gas Barrier	Automotive Fuel Line Parts, Gas/Oil Pipeline Coatings
Lubrication	Anti-Wear Low Friction	Engine Oil, Gear Oil, Machine Oil
Acid/Chemical Resistance	Anti-Corrosion	Industrial Parts Coatings Oil Drilling Pipe Coating

XG TIM™ Thermal Interface Materials

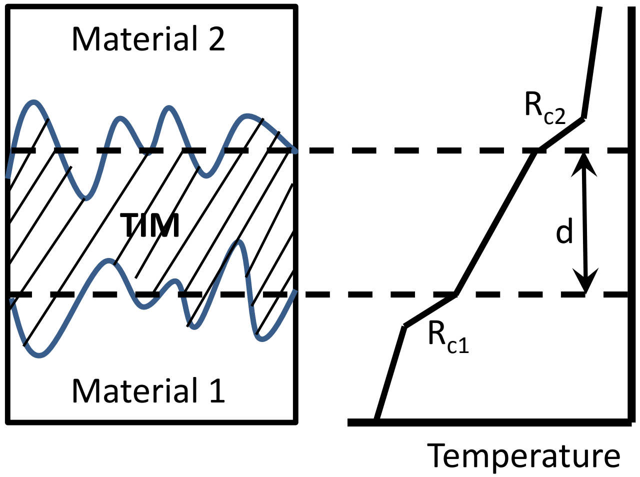
Requirements of Thermal Interface Materials (TIMs)

- TIMs enable efficient heat transfer from device to chip package / PCB to heat sink
- Design goal is to minimize thermal impedance and keep device T_j below a specified limit, e.g. 100°C



<u>TIM Requirements:</u>

- High thermal conductivity (k)
- Minimum contact resistance ($R_{contact}$)
  - Easily deformed at low pressures to fill voids
  - Good wetting to the interfacial surfaces
- Minimum bond-line thickness (d)
- Low cost
- Ease of application – pick-&-place, dispense
- High reliability – out-gassing, end-of-life



$$\text{Thermal Impedance, } \theta \ (\mathrm{cm^2 \cdot K/W}) = R_{contact\,1} + \frac{d\ (\mathrm{mm})}{k\ (\mathrm{W/m \cdot K})} + R_{contact\,2}$$

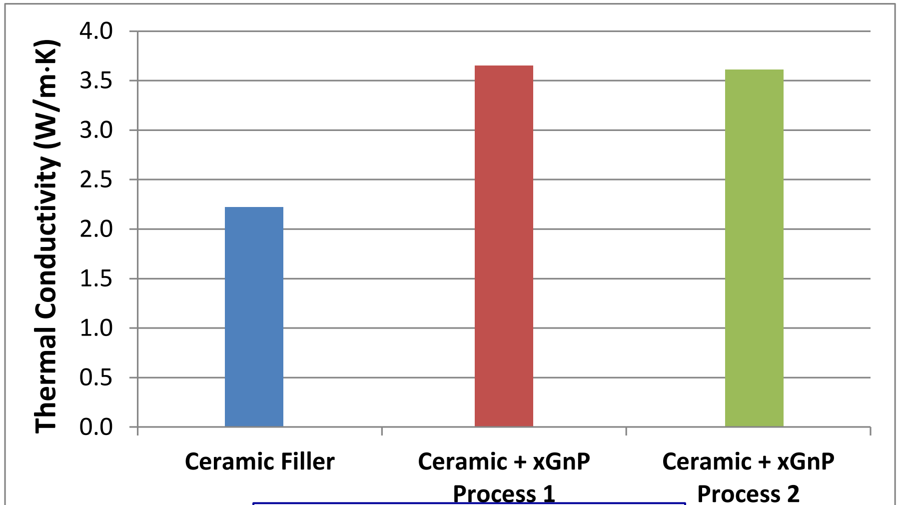
XG TIM™ GR Thermal Grease: High Performance, Cost Effective

- XG TIM™ thermal grease is composed of xGnP® graphene nanoplatelets and ceramic particles dispersed in silicone oil
- It flows easily under relatively low pressure (≈20psi) to achieve thin bond line (<100μm) and low thermal impedance at the interface
- Low xGnP® loading levels and economical ceramic fillers result in the best price-performance

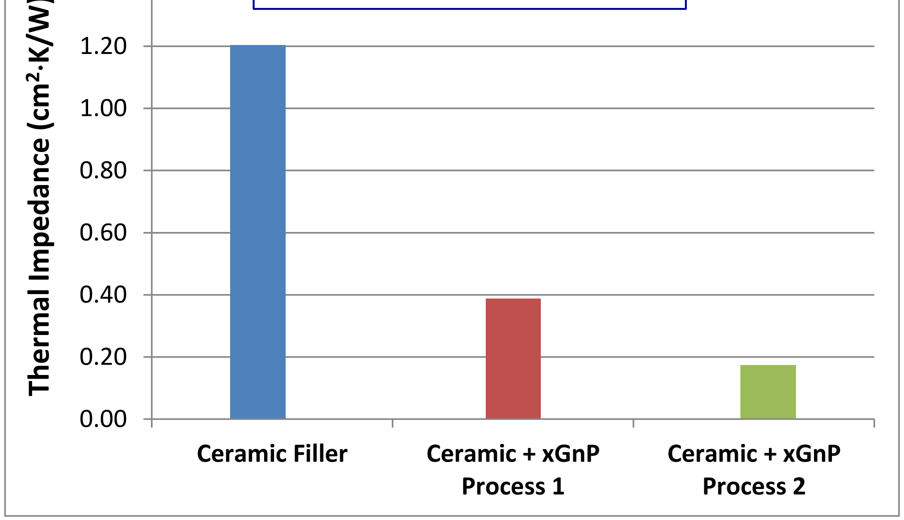
Particle	Intrinsic Thermal Conductivity (W/m·K)
CNT	> 1,000
Graphene Nanoplatelet	> 1,000
Diamond	> 1,000
Silver	410
Copper	360
Boron Nitride	300
Aluminum	205
Natural Graphite	200
Aluminum Oxide	30
Zinc Oxide	20

Enhanced Thermal Conductivity with xGnP® Nanoplatelets

- xGnP® nanoplatelets provide enhanced particle-to-particle heat conduction compared to ceramic fillers only → higher thermal conductivity
- xGnP® nanoplatelets enable lower bond-line thickness at the same contact pressure → lower thermal impedance
- xGnP® nanoplatelets provide significant improvement in thermal performance at low loading levels → cost effectiveness



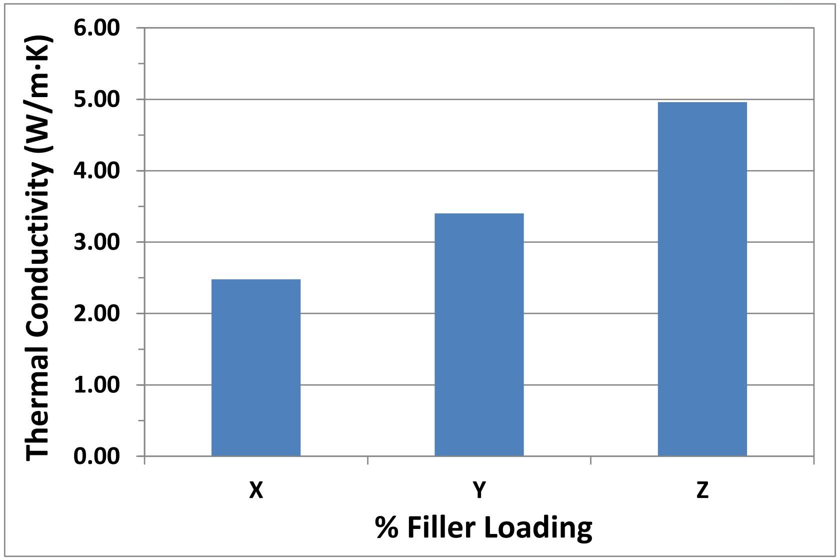
Low xGnP® loading

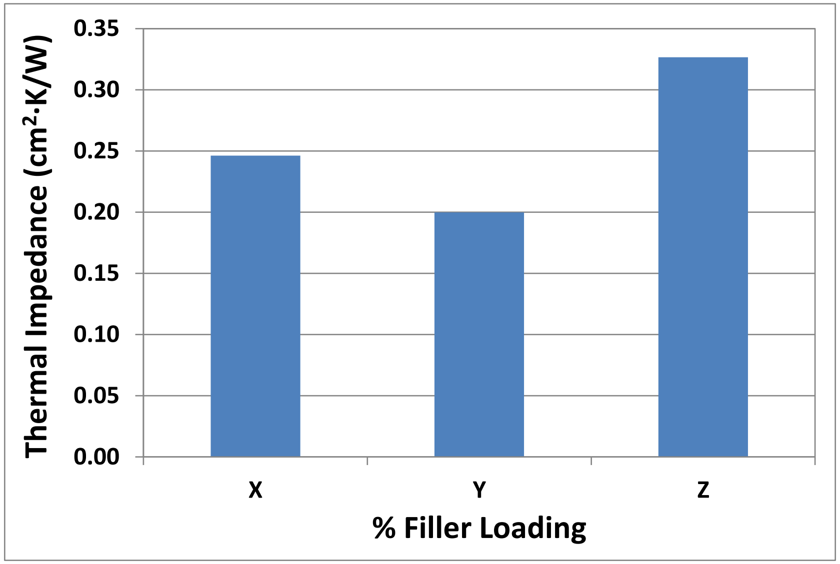


All thermal measurements at 20psi using Laser Flash Analysis (ASTM E1461)

Optimization of Filler Content

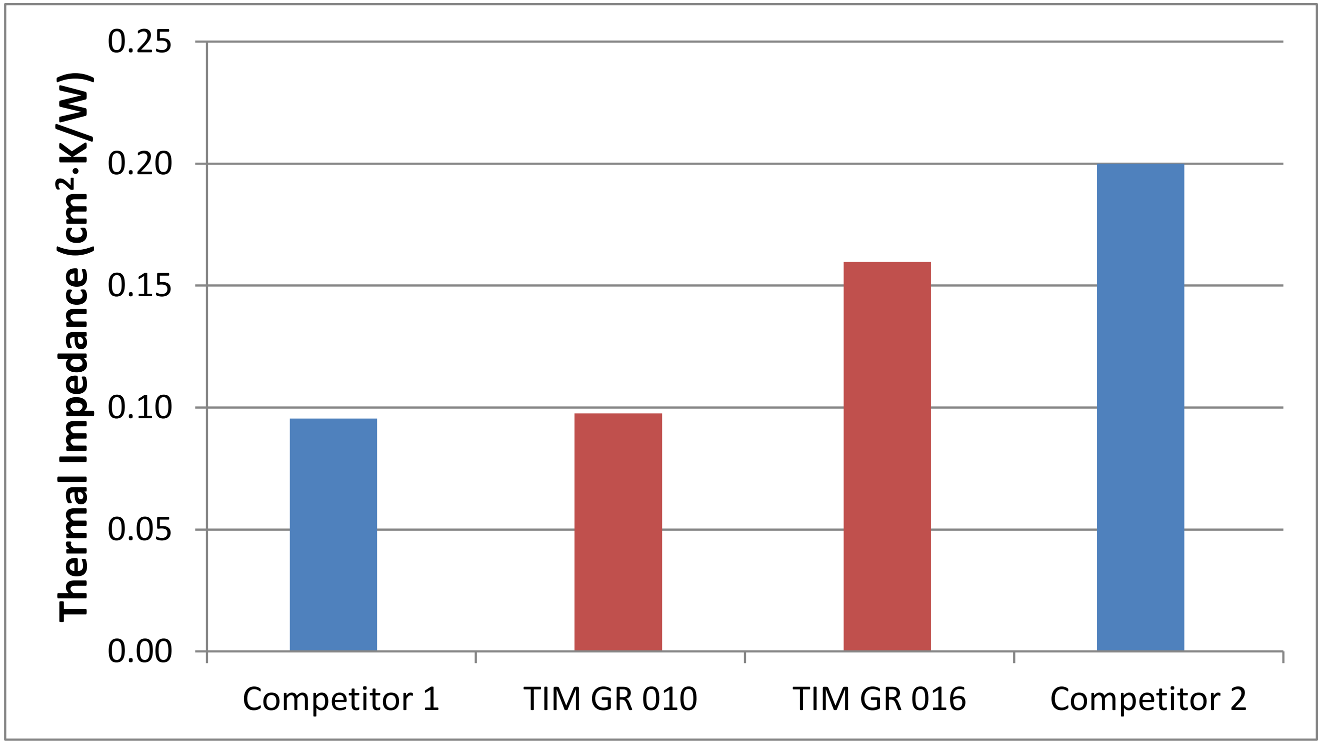
- In general, thermal conductivity increases with increasing % filler loading
- However, higher filler content can also increase the TIM viscosity which can:
  - Make it harder to apply
  - Increase the bond-line thickness which offsets the higher conductivity
- The result is an optimum % filler loading for minimum thermal impedance





All thermal measurements at 20psi using Laser Flash Analysis (ASTM E1461)

Competitive Performance of XG TIM™ GR Thermal Grease



XG TIM™ GR formulations demonstrated comparable thermal impedances to some of the best available commercial greases

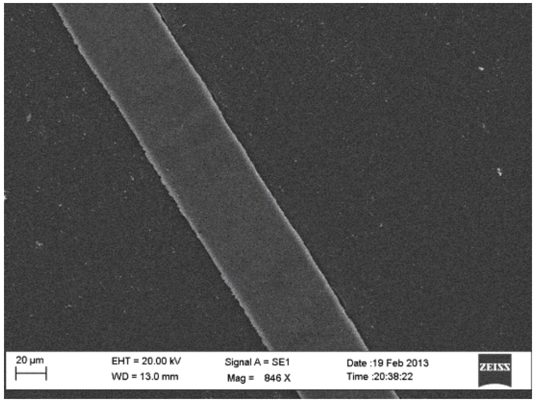
All thermal measurements at 20psi using
Laser Flash Analysis (ASTM E1461)

XG Leaf® Heat Spreaders

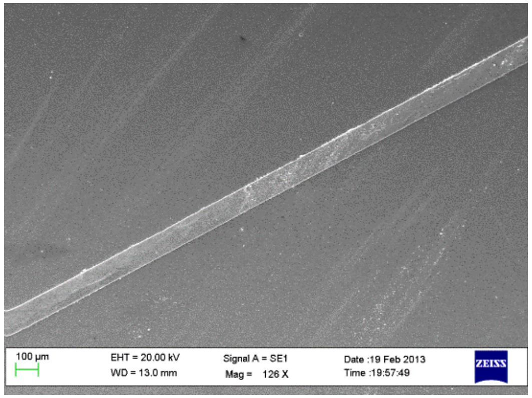
XG Leaf® – Superior Properties

Thin, flexible, lightweight sheets formed from xGnP® graphene nanoplatelets

- Excellent in-plane Thermal Conductivity: > 500 W/m·K
- Low through-plane Thermal Conductivity: 3 W/m ·K
- Available in various Thickness: 50, 120, 240 µm
- Customizable Mechanical Properties
- Bending Flexibility
- Light Weight
- Corrosion Resistant
- Excellent in-plane Electrical Conductivity : > 3000 S/cm
- Excellent EMI Shielding: > 50 dB (30MHz to 1.5GHz)



50µm XG Leaf®



120µm XG Leaf®

XG Leaf® – Competitive Positioning

XG Leaf® vs graphite foil:

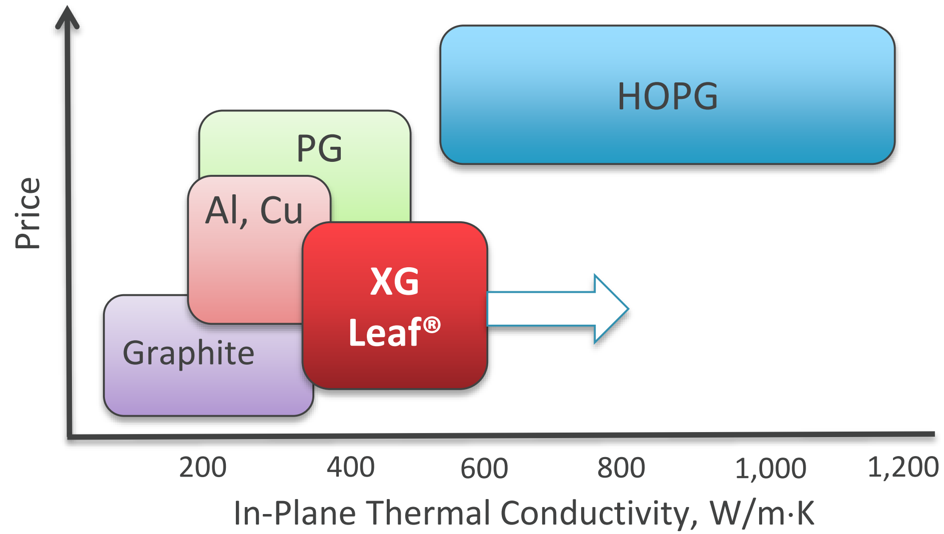
- Better thermal and electrical properties
- Tailored compositions of graphene nanoplatelets and selected additives or coatings for specific properties
- Better 2-D anisotropic properties

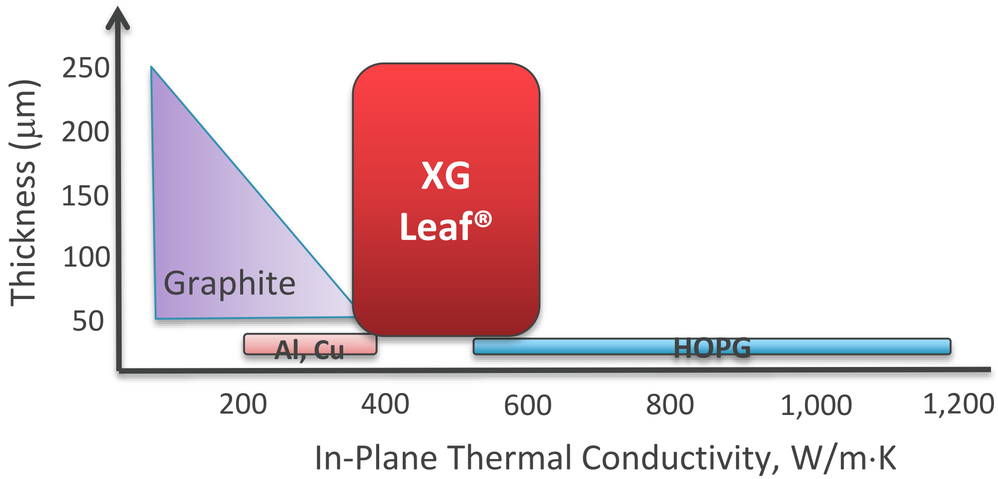
XG Leaf® vs HOPG:

- Available in broad thickness range from ~30 to 240 microns
- Thermal conductivity invariant with thickness
- Lower cost

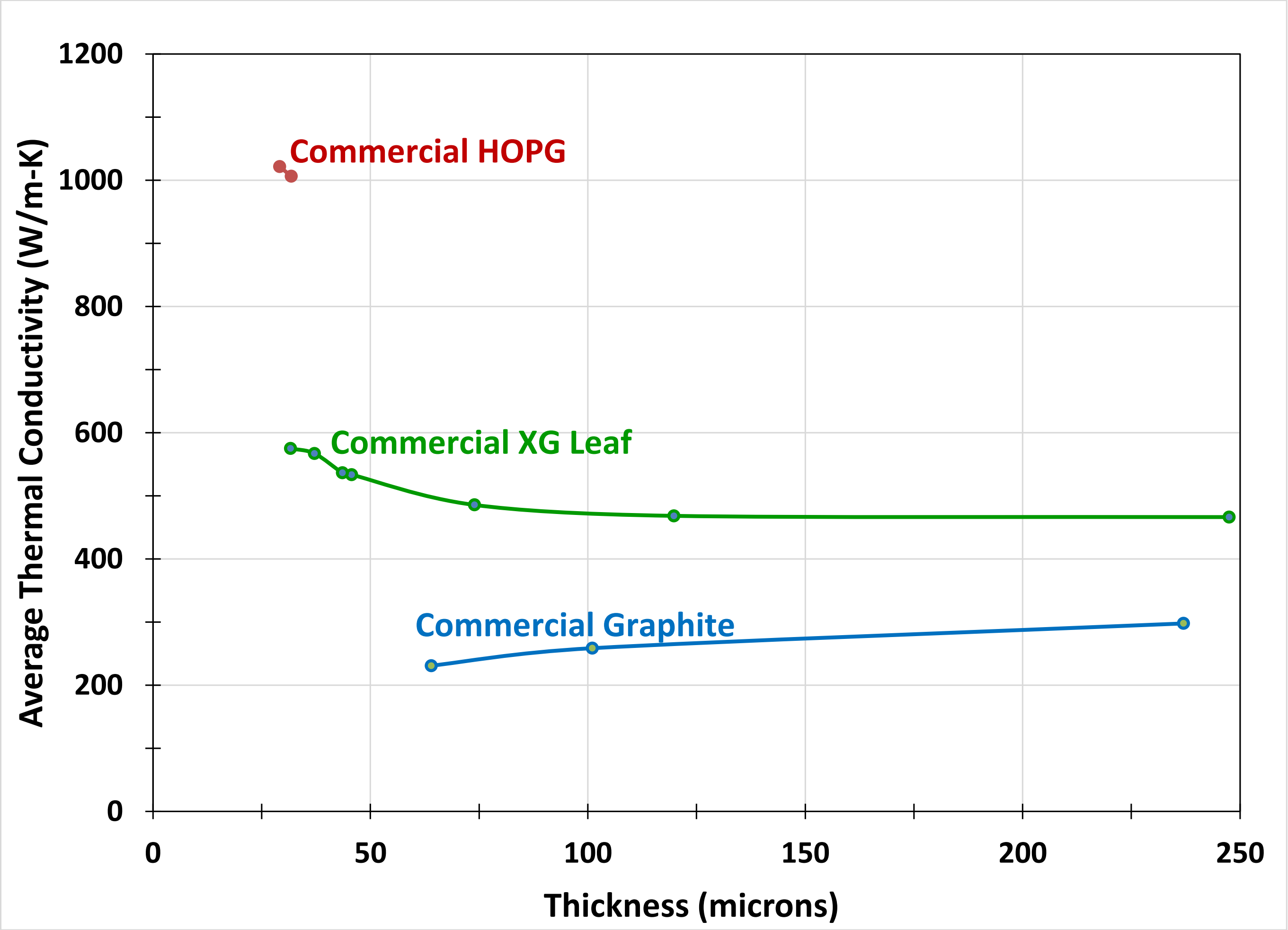
XG Leaf® vs metal foil:

- Lighter, non corrosive, better chemical resistance
- Better thermal properties
- Much better two-dimensional anisotropic properties





Comparison of Thermal Conductivities

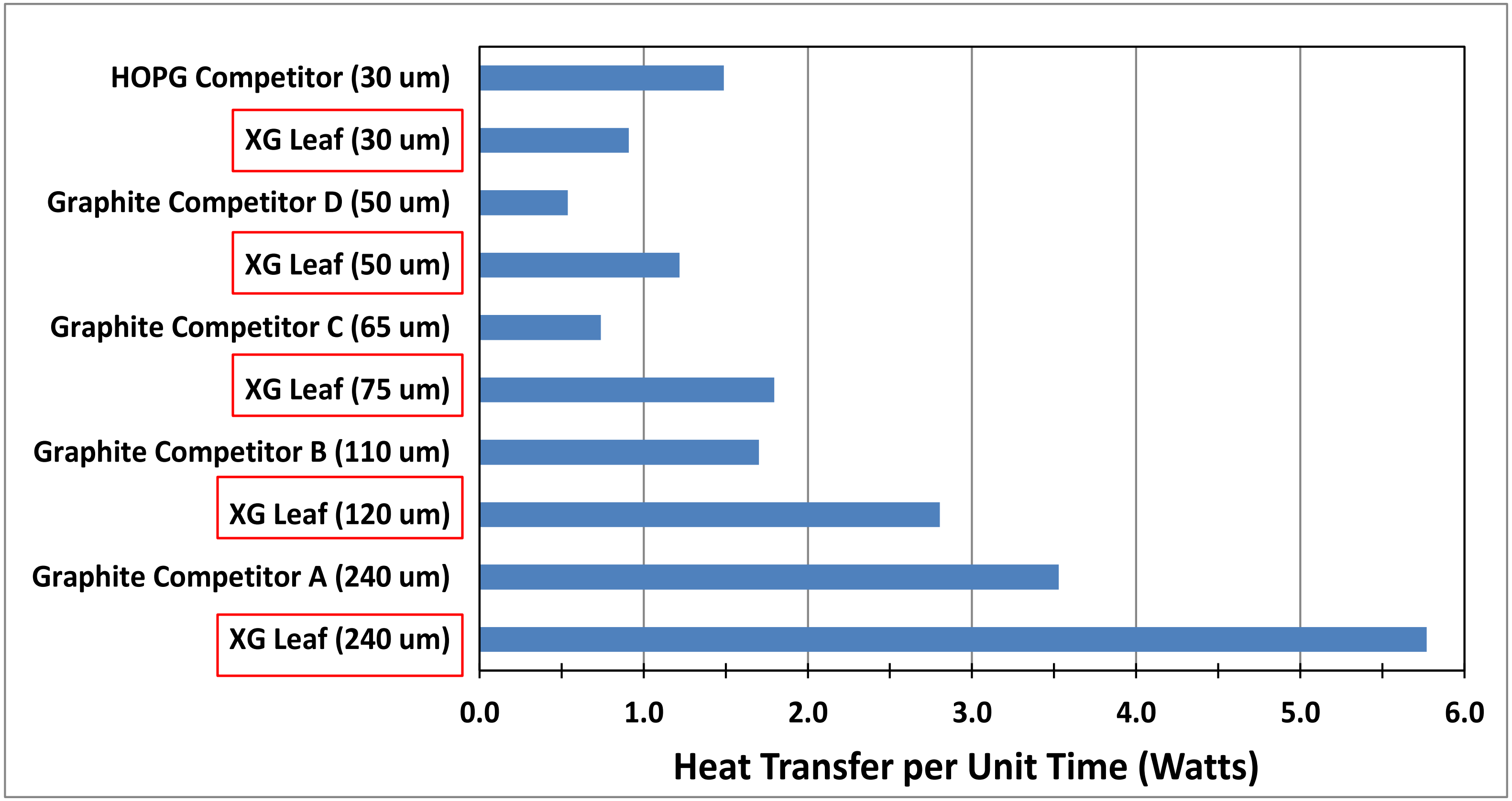


All thermal measurements using Laser Flash Analysis (ASTM E1461)

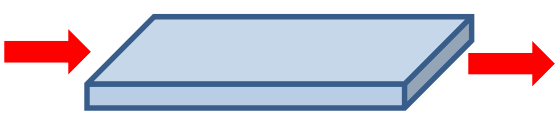
- **Thermal conductivity data of XG Leaf® as a function of thickness, compared to other commercial products, measured on the same tester**
- **Caution using datasheets for comparison, e.g. above HOPG was specified at 1600 W/m·K**

Comparison of Heat Transfer Capabilities

Superior heat transfer capability of thicker XG Leaf® compared to Graphite and higher conductivity HOPG



All thermal measurements using Laser Flash Analysis (ASTM E1461)



Assume width=10cm, length=10cm, $\Delta T=50^oC$

$$Q\ (W) = \frac{k(W/m\cdot K) \times w(cm) \times t(cm) \times \Delta T\ (K)}{l(cm) \times 100(cm/m)}$$

XG Leaf® – Tailoring for different Applications

- Multi-function solutions, e.g. XG Leaf® for heat spreading plus EMI Shielding in consumer electronics
- Cooling Li-ion battery packs
- XG Leaf® formulation for resistive heating in automotive, industrial, medical applications
- Coatings on XG Leaf® sheets
  - Adhesive (including thermally conducting)
  - Abrasion resistance
  - Tensile strength
  - Electrically insulating
- Lamination
- Die cut

XG Ink™ Inks and Coatings

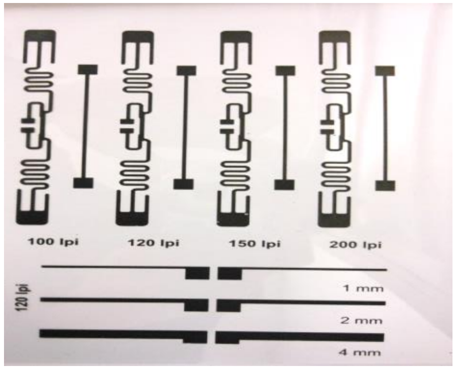
Electrically Conductive Inks

Portfolio

- Water-Based Gravure Ink
  - $5 - 10\ \Omega/\square/\text{mil}$
  - Viscosity : ~ 200 cP
- Water-Based Flexo Ink
  - $5 - 10\ \Omega/\square/\text{mil}$
  - Viscosity : ~ 400 cP
- Water-Based Offset Ink
  - $5 - 10\ \Omega/\square/\text{mil}$
  - Viscosity : ~ 5,000 cP
- Solvent-Based Gravure Ink
  - $10 - 15\ \Omega/\square/\text{mil}$
  - Viscosity : ~ 200 cP

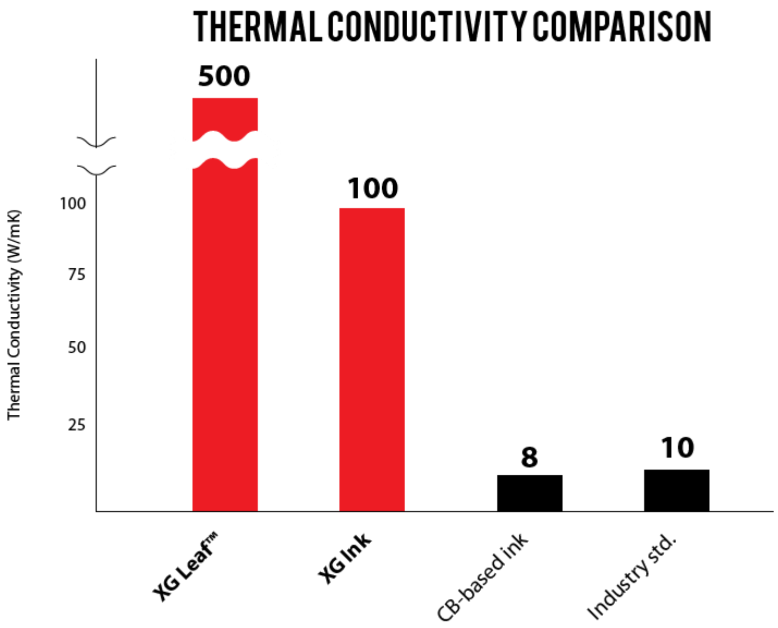
Advantages

- Lower cost compared to Ag inks
  - Replace/reduce Ag consumption
- Higher conductivity compared to Carbon Inks
- Good print quality
- Water-based formulations available



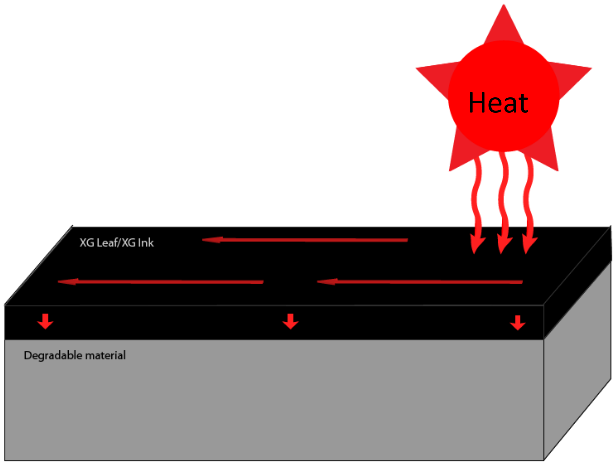
Other Ink formulations in progress, driven by customer requirements, e.g. lower resistivity inks, solvent-based screen-printed inks, etc.

Thermally Conductive Coatings



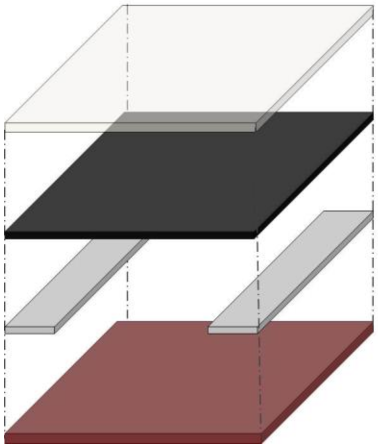
Water Based Thermal Spray Coating

- Thermal Conductivity: 5 – 10 W/m·K (through-plane)
- Thermal Conductivity: 50 – 100 W/m·K (in-plane, compressed)
- Electrically Conductivity: 1 – 5 Ω/☐/mil
- Viscosity : ~ 100 cP

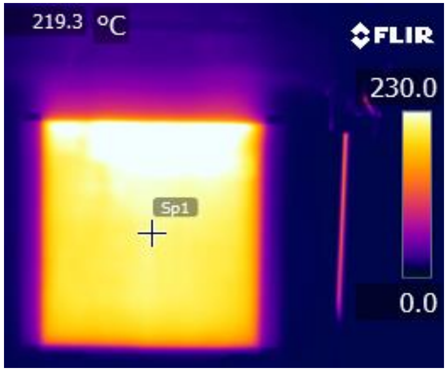


XG Ink™ can be painted onto most surfaces and tailored for adhesion and durability

Resistive Heating Paste



- Excellent Energy Efficiency
  - Low Power => Energy Savings
  - Low AC/DC voltage
  - 200℃ heating with below 12V DC
- High Heating Uniformity
- Good Heating Stability
- Fast Ramp Rates
- Application Flexibilty



Characteristic	Performance
Thermal stability (℃)	250
Resistivity (Ω·cm)	2.64×10^{-2}
Sheet resistance (Ω/□)	5~100
Curing temp. (℃) / time (min)	135 / 3~5
Substrate	PI, Fabric, Ceramic, Plastic, Mica, etc.
Heat-up rate (℃/min)	80

Summary

- **XG Sciences**
  - Recognized market leader in Graphene technology and business execution (Lux Research, July 2015)
  - World class strategic partners and licensees
  - Significant IP position
  - Low cost, scalable manufacturing capabilities

- **XG TIM™ Thermal Interface Materials**
  - xGnP® graphene nanoplatelets provide enhanced heat conduction at low loading levels
  - XG TIM GR thermal grease formulations demonstrated comparable thermal impedance to the best commercial greases, ≈ 0.1 $cm^2 \cdot K/W$ – opportunities for further improvement
  - TIM platform extension to adhesives, pads, PCMs, ….
  - High Performance, Cost Effective

- **XG Leaf® Heat Spreaders**
  - Better thermals, >500 $W/m \cdot K$ compared to graphite, metal foils, and lower cost compared to HOPG
  - Flexibility of available thickness ≈ 30 to 240 μm to fit the application
  - Tailored compositions and selected coatings designed for specific properties
  - High Performance, Cost Effective

- **XG Ink ™ Inks and Coatings**
  - Better conductivity compared to carbon inks, and lower cost compared to Ag inks
  - Energy efficient resistive heating

Thank You

Percy Chinoy, Ph.D.

Sr. Director, Business Development

+1 517 999 5452

p.chinoy@xgsciences.com